Paleomagnetism of the Abitibi dyke swarm, southern Superior Province, and implications for the Logan Loop[1]

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The trace of the Middle Proterozoic Logan Loop of the North American apparent polar wander path is controversial. The older 1270–1110 Ma limb of the loop is poorly constrained, while the depth of the loop, based on the 1110–1080 Ma Keweenawan data of the younger limb, is thought by some to be largely an artifact of reversal asymmetry in the Earth's magnetic field. Paleomagnetism of the 1141 Ma Abitibi mafic dyke swarm is one of the keys to constraining the geometry of the Logan Loop.

Unfortunately, previous paleomagnetic studies failed to distinguish dykes of the northeast-trending Middle Proterozoic olivine-bearing Abitibi swarm from subparallel Early Proterozoic olivine-free Biscotasing (formerly Preissac) dykes, and hence paleomagnetic poles determined in these studies should no longer be used. In the present study, sampling of eight Abitibi dykes has identified 23 normally magnetized sites in four dykes and, for the first time, five reversely magnetized sites in three dykes. One of the normally magnetized sites corresponds to the locality for which a high-precision U–Pb age was previously reported. A baked contact test establishes that the characteristic remanence of one normally magnetized dyke is primary. In addition, sites along individual dykes exhibit much smaller secular variation than is observed between dykes, indicating that the remanences of the other dykes are also primary. One of the normally magnetized dykes, the 700 km long Great Abitibi dyke, exhibits two primary directions that correspond to two geochemically distinct magma pulses. The five normally magnetized units, which consist of four separate dykes plus the second pulse of the Great Abitibi dyke, yield a well-defined mean paleomagnetic pole at 42.8°N, 151.5°W, dm = 16.3°, dp = 12.5°. It falls close to the reversely magnetized poles from the Keweenawan Track and establishes a minimum depth for the Logan Loop of about 40°. The reversed-polarity data from three other dykes are more scattered and may not average out secular variation. Hence, the present study is inconclusive regarding asymmetry of the Earth's magnetic field at 1141 Ma, even though a mean pole based on combined normal- and reversed-polarity dykes is indistinguishable from that based on normal-polarity dykes alone.

Le tracé de la boucle de Logan, lequel correspond au segment du Protérozoïque moyen de la courbe de la dérive des pôles apparente de l'Amérique du Nord, fait l'objet d'une controverse. La partie la plus ancienne de la boucle, 1270–1110 Ma, est pauvrement délimitée, et la profondeur de la boucle est estimée à partir du segment le plus jeune de la boucle, 1110–1080 Ma, correspondant au Keweenawan, et que certains auteurs attribuent à un artefact créé en bonne partie par une asymétrie dans l'inversion du champ magnétique terrestre. Le paléomagnétisme de l'essaim de dykes mafiques de l'Abitibi, daté de 1141 Ma, constitue une des clés permettant de définir la géométrie de la boucle de Logan.

Malheureusement, les études paléomagnétiques réalisées à ce jour n'ont pas réussi à faire la distinction des dykes à olivine de l'essaim d'Abitibi, de direction nord-est, datés du Protérozoïque moyen, d'avec les dykes dépourvus d'olivine de Biscatasing (anciennement Preissac), d'orientation subparallèle, datés du Protérozoïque précoce et, par conséquent, il ne faut plus utiliser les pôles paléomagnétiques qui ont été déterminés par ces études. Ici, un échantillonnage de huit dykes d'Abitibi a permis de repérer dans quatre dykes 23 sites d'aimantation normale, et, pour la première fois, cinq sites d'aimantation inversée dans trois dykes. Un des sites d'aimantation normale coïncide avec une localité qui avait été datée par U–Pb avec grande précision. Un test de contact de la roche cuite a établi que la rémanence caractéristique d'un dyke aimanté normalement est primaire. En outre, les sites le long des dykes individuels exhibent une variation séculaire qui est moins importante que celle observée entre les dykes, ce qui confirme que les rémanences des autres dykes sont éaglement primaires. Un des dykes aimantés normalement, le dyke du Grand Abitibi d'une longueur de 700 km, fournit deux directions primaires qui correspondent à deux pulsations magmatiques géochimiquement distinctes. Les cinq unités normalement aimantées, incluant quatre dykes séparés plus la seconde pulsation du dyke du Grand Abitibi, donnent un pôle magnétique moyen bien défini à 42,8°N, 151,5°W, dm = 16,3°, dp = 12,5°. Ce pôle tombe près des pôles d'aimantation inversée du segment Keweenawan, et détermine une profondeur d'écart minimum pour la boucle de Logan d'environ 40°. Les données de polarité inversée sont plus dispersées dans les trois autres dykes, et elles ne peuvent fournir une valeur moyenne de variation séculaire. Donc, la présente étude démontre l'existence d'une asymétrie dans le champ magnétique terrestre il y a 1141 Ma, même si un pôle moyen déduit des dykes de polarités normale et inversée ne peut être distingué de celui qui dérive uniquement des dykes de polarité normale.
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Introduction

An important area of paleomagnetic research involves the study of mafic dykes (Buchan and Halls 1990) because they usually carry a stable remanence and can now be readily dated with great precision using the U–Pb technique on trace amounts of baddeleyite or zircon (Krogh et al. 1987). Buchan et al. (1993*b*) have further demonstrated that paleomagnetic poles can be dated with high precision using the methodology of sampling for paleomagnetism and U–Pb geochronology at the same localities and by conducting rigorous paleomagnetic field tests to establish that the remanences are primary.

The northeast-trending Abitibi dyke swarm of the southern

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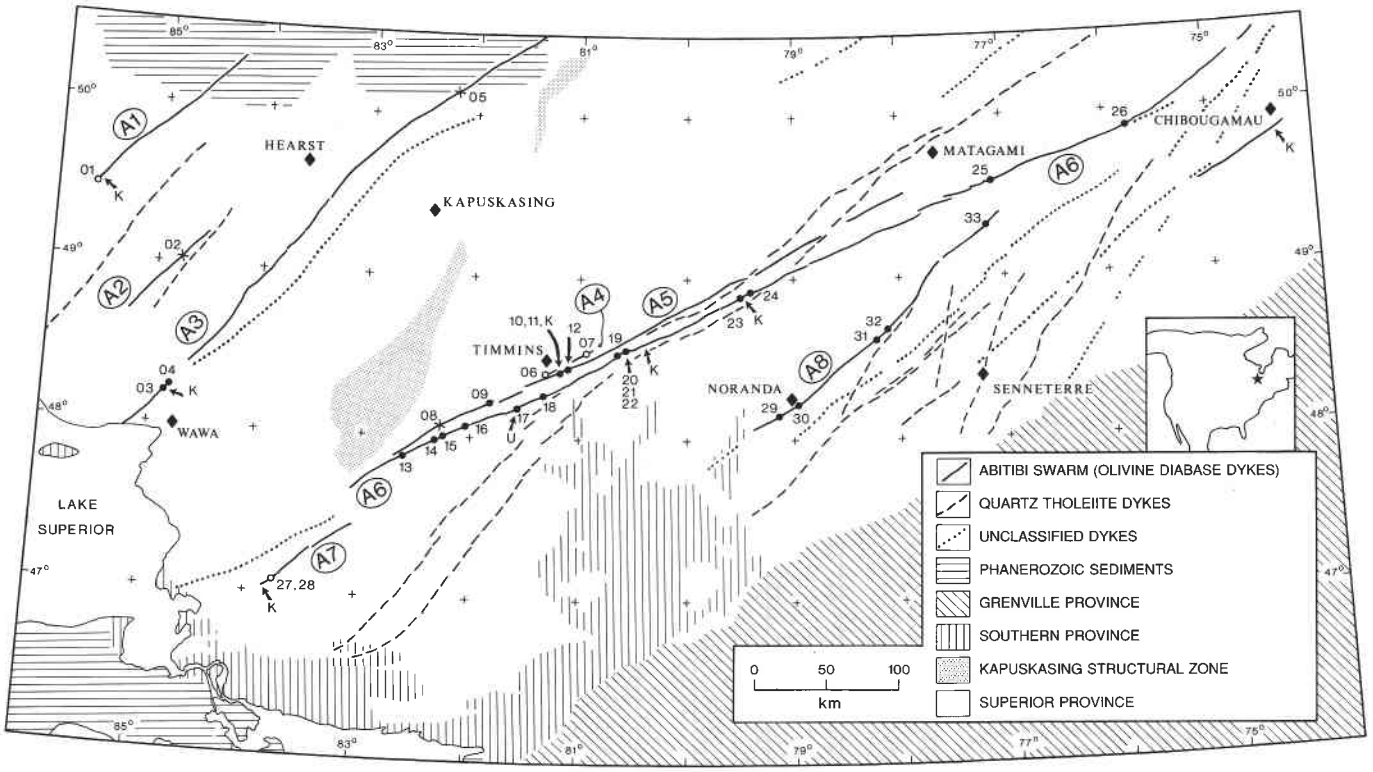


FIG. 1. Northeast-trending dykes in the southern Superior Province. Paleomagnetic sampling sites (labelled 01–33) were located in dykes (labelled A1–A8) of the Abitibi dyke swarm. Sites shown as closed (open) circles carry normal (reversed) characteristic magnetization directions. Sites shown as crosses have unstable or scattered magnetization directions. K and U, K–Ar biotite and U–Pb baddeleyite radiometric dating sites, respectively (Table 1).

Superior Province (Fig. 1) represents a key target for paleomagnetic study for several reasons. Firstly, it has already been precisely dated at 1140.6 ± 2.0 Ma (Krogh et al. 1987). Secondly, its pole position is important in delineating the depth and form of the Middle Proterozoic Logan Loop (Robertson and Fahrig 1971) of the North American apparent polar wander path. Thirdly, a multidisciplinary study involving paleomagnetism, geochemistry, petrography, and anisotropy of magnetic susceptibility of the largest dyke in the Abitibi swarm, the Great Abitibi dyke, has recently been completed (Ernst et al. 1987; Ernst 1989, 1990; Ernst and Bell 1992).

Unfortunately a reliable pole for the Abitibi swarm has not yet been established because of confusion in distinguishing northeast-trending dykes of widely different ages in the southern Superior Province. There have been numerous studies of the paleomagnetism, geochronology, or petrography of these dykes during the past three decades (e.g., Strangway 1961; Fahrig et al. 1965; Larochelle 1966; Gates and Hurley 1973; Irving and Naldrett 1977; Hanes and York 1979; Buchan and Schwarz 1981; Schutts and Dunlop 1981). Larochelle (1966) recognized three distinct periods of dyke emplacement based on paleomagnetism. However, in most of the studies and in reviews (e.g., Irving and McGlynn 1981; Roy 1983; Windley 1984; Dunsmore and Symons 1990) the dykes were considered to belong to a single swarm and usually labelled as Abitibi dykes. The Rb–Sr age of about 2150 Ma reported by Gates and Hurley (1973) for Abitibi dykes is now known to be derived from at least three northeast-trending dyke swarms of vastly different ages.

W.F. Fahrig in Goodwin et al. (1972), Card (1981), and Fahrig (1987) summarized the evidence for an Early Proterozoic quartz tholeiite swarm and a Middle Proterozoic olivine diabase swarm (Fig. 1), both of northeast trend, and reserved the name Preissac for the former and Abitibi for the latter. As noted above, the Abitibi swarm has been dated at 1141 Ma (Krogh et al. 1987). Preissac dykes have recently been subdivided on the basis of paleomagnetism and U–Pb geochronology (Buchan et al. 1993*a*) into a 2167 + 1.4 Ma Biscotasing swarm, a 2214 + 13 Ma Senneterre swarm, and a third set that is undated but may belong to the Matachewan swarm of age 2452 + 3 Ma (Heaman and Tarney 1989). Biscotasing dykes carry primary magnetizations that cannot be readily distinguished from and have often been averaged with those from the Abitibi dykes. In addition, northeast-trending Kapuskasing dykes, with an Ar^{40}/Ar^{39} age of 2043 ± 14 Ma (Queen et al. 1989), have been reported from the southern Kapuskasing Structural Zone (KSZ), but are not shown in Fig. 1 because of their limited areal extent.

Abitibi swarm

The Abitibi swarm of the southern Superior Province comprises a number of widely spaced olivine gabbro dykes that trend northeast and can, in some cases, be traced for hundreds of kilometres (Fig. 1). The longest, the Great Abitibi dyke (dyke A6), can be followed for 700 km (Ernst 1991; Ernst and Bell 1992). Abitibi dykes are nearly vertical and vary in thickness from 15 to 250 m.

The U/Pb baddeleyite age of 1141 (Krogh et al. 1987) was obtained at site 17 of dyke A6 (Fig. 1). Much less precise K/Ar biotite ages that fall in the 1220–1035 Ma range have been reported for five of the eight northeast-trending dykes

TABLE 1. U–Pb baddeleyite and K–Ar biotite radiometric ages for Abitibi dykes

Dyke[a]	Radiometric age[b] (Ma)	Reference
A1	1125±40 (K)	Wanless et al. 1966, p. 66
A3	1120±65 (K)	Leech et al. 1963, pp. 73–74
		Wanless et al. 1965, p. 91
A5	1220±40 (K)	Wanless et al. 1965, p. 92
A6	1140.6±2 (U)	Krogh et al. 1987
	1240±50 (K)	Wanless et al. 1965, pp. 86–87
	1092±39 (K)	Wanless et al. 1968, p. 108
A7	1035±35 (K)	Wanless et al. 1968, p. 96

[a]For dyke locations see Fig. 1. For dyke A6, the 1240 Ma age is located near sites 20–22 and the 1092 Ma age is located between sites 23 and 24.
[b]For dating locations see Fig. 1. K, K–Ar biotite age; U, U–Pb baddeleyite age. Uncertainties quoted at the 2σ level.

studied in this paper (Table 1). The dating indicates that all are of Middle Proterozoic rather than Early Proterozoic age and likely belong to the Abitibi swarm. It is assumed that all dykes of the Abitibi swarm were emplaced within a few million years as appears to have been the case with other Proterozoic dyke swarms for which precise dating has been carried out (e.g., LeCheminant and Heaman 1989).

Abitibi dykes are distinct in their chemical and petrographic characteristics from the Early Proterozoic northeast-trending quartz tholeiite dykes of the Biscotasing and Senneterre swarms (W.F. Fahrig in Goodwin et al. 1972; Card 1981; Condie et al. 1987; Ernst 1989; Ernst and Bell 1992; Buchan et al. 1993*a*; this study). Essential differences can be summarized as follows: Abitibi dykes are olivine normative, olivine-bearing except in some parts of the evolved interiors of wide dykes, and generally contain a few percent apatite. They plot in the alkaline field on alkali–silica diagrams, and are light rare-earth element enriched. Conversely, the older quartz tholeiite dykes of the Biscotasing and Senneterre swarms are quartz normative, olivine free, contain only trace apatite, plot in the subalkaline field on alkali–silica diagrams, and are less light rare earth enriched.

In the present study, petrographic examination of a representative set of thin sections from along each Abitibi dyke was conducted. Two petrographic types were identified. Each dyke except perhaps for dyke A1, as discussed below, is mainly composed of olivine gabbro. Some dykes also have interior units which on the basis of mineral proportions should be termed an olivine-bearing monzogabbro, but which could alternatively be termed an olivine-bearing monzodiorite on the basis of having plagioclase compositions of An $<$ 50 (classifications after Streckeissen 1976). In particular, dyke A6 contains an interior monzogabbro unit extending from sites 16 to 24 inclusive, and at these sites the contact between the interior unit and the marginal olivine gabbro is gradational (Ernst et al. 1987; Ernst 1989; Ernst and Bell 1992). On the basis of more limited study at sites 11 and 12 we think that dyke A5 also contains both olivine gabbro and the interior monzogabbro unit. Only the monzogabbro unit was observed in site 01 of dyke A1, but this site had poor outcrop exposure and until more sites have been studied, the absence of olivine gabbro in this dyke remains speculative. All the dykes except A8 contain abundant apatite, which averages 3% with a maximum of 7%; dyke A8 contains only trace apatite. All dykes contain Fe–Ti oxides, consisting of composite grains of titanomagnetite and exsolved ilmenite as well as separate ilmenite grains. In dyke A6 these have microprobe-analyzed compositions of Usp_{32-74} and Ilm_{93-97} (for both exsolved ilmenite as well as the separate ilmenite grains). The presence of orange–red biotite and pinkish augite indicate high Ti contents; biotites from dyke A6 have high Ti contents of 6.0 wt.%, and augites from dykes A3 and A6 have high Ti contents of between 1 and 2%. An unusual feature of dyke A2 at site 02 is the presence of abundant phenocrysts of colourless fresh plagioclase up to 2 cm in size near the southern margin of the dyke. Generally the dykes are unmetamorphosed and unaltered, although some alteration is observed locally (e.g., from dyke A1, site 01).

Samples could not be obtained from four other probable Abitibi dykes, but their locations are given for future study. They include (*i*) an olivine gabbro dyke located 12.5 km due south of Chibougamau on Lake Chibougamau (Fig. 1) which has a K/Ar biotite age of 1035 ± 35 Ma (Wanless et al. 1970, pp. 67 and 68); (*ii*) an olivine gabbro dyke, not shown, that cuts the 1888 Ma (L.M. Heaman, personal communication, 1991) Cargill complex (Allen 1972) at approximately 49.3°N, 82.8°W; (*iii*) a dyke located north of Fig. 1 at approximately 51.6°N, 77.4°W with Abitibi-like chemistry and petrography (sample 34 in Condie et al. 1987); and (*iv*) a 1 m wide amygdaloidal olivine-bearing dyke, not shown in Fig. 1, with northeast trend on Shawmere Lake at approximately 48.3°N, 82.6°W in the southern KSZ (H. Halls, personal communication, 1992). Other Abitibi dykes may be located in the KSZ (Percival 1981; J.A. Percival, personal communication, 1992). However, a dyke from the southern KSZ, having trace olivine and suggested as a possible Abitibi dyke by Halls and Palmer (1990, site 181), is now interpreted as belonging to the Kapuskasing swarm (H. Halls, personal communication, 1992).

Field and laboratory procedure

This paleomagnetic study is based on data from 33 sampling sites along the eight Abitibi dykes described above. Oriented blocks or drill cores were collected from each site. Two or three cylindrical specimens with a 2.5 cm diameter and 2.2 cm length were cut from each sample and their magnetization measured using a Schonstedt spinner magnetometer. Typically, two specimens per sample were demagnetized in about 15 steps to peak alternating fields of 100 mT with a Schonstedt single-axis demagnetizer. Samples from selected sites were thermally cleaned using a Schonstedt thermal demagnetizer modified to maintain temperatures within ±5°C. At site 17, selected samples were thermally demagnetized in nine steps to 580°C; samples from other sites were thermally demagnetized in either 18 or 19 steps to 580°C.

Stable remanence directions for individual specimens were calculated using the least squares fit method of Kirschvink (1980) based on linear segments identified on orthogonal component diagrams. Specimen directions were then averaged at the sample, site, dyke, and swarm level. To facilitate direct comparison of data (Table 2) from the broad geographic distribution of Abitibi dyke site localities (Fig. 1), site directions have been recalculated for a common reference locality (48°N, 81°W).

Paleomagnetic results

At 26 of the 33 sampling sites, stable remanence directions were readily determined using alternating-field (AF) or thermal demagnetization. These stable sites carry either a reversed or

TABLE 2. Summary of characteristic remanences for Abitibi dyke sites

Site No.	Dyke	N_A (n_A)	N_T (n_T)	Lat. (°)	Long. (°)	Thickness (m)	Dip	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)
01	A1	10 (4)	8 (0)	49.459	85.625	~90	—	95.3	−49.0	99.3	−45.2	30	17.1
02	A2	15 (0)	1 (0)	49.022	84.772	>15	85°S	—	—	—	—	—	—
03	A3	7 (0)	7 (6)	48.205	84.845	~50	85°N	305.0	58.5	306.8	56.4	230	4.4
04	A3	7 (0)	7 (6)	48.233	84.817	>30	—	288.7	50.8	291.2	48.0	98	6.8
05	A3	9 (0)	10 (0)	50.145	82.211	~50	80°S	—	—	—	—	—	—
06	A4	7 (7)	1 (0)	48.387	81.298	>35	—	124.7	−71.4	125.2	−71.2	129	5.3
07	A4	7 (7)	1 (0)	48.533	80.937	—	—	127.1	−72.0	127.8	−71.8	154	4.9
08	A5	5 (0)	—	48.079	82.290	—	—	—	—	—	—	—	—
09	A5	5 (5)	1 (0)	48.218	81.832	—	—	313.6	49.1	314.1	48.5	131	6.7
10	A5	6 (5)	—	48.415	81.202	>30	—	316.7	66.5	317.1	66.2	42	11.9
11	A5	10 (8)	—	48.429	81.175	—	—	304.3	54.6	304.6	54.3	55	7.6
12	A5	7 (7)	—	48.441	81.117	>80	—	306.0	67.2	306.5	66.9	42	9.5
13	A6, unit 1	10 (8)	—	47.878	82.620	89	85°N	275.0	65.2	275.9	64.4	115	5.2
14	A6, unit 1	9 (9)	—	47.978	82.325	149	70°N	292.4	53.0	293.2	52.2	59	6.8
15	A6, unit 1	12 (12)	—	48.002	82.250	138	75°N	288.5	57.5	289.2	56.7	82	4.8
16	A6, unit 2	1 (1)	—	48.072	82.052	109	—	311.3	63.5	311.7	63.1	—	—
		4 (4) B	—					310.0	66.3	310.3	65.8	125	8.3
		3 (3) U	—					218.6	−15.7	219.3	−16.4	47	18.2
17	A6, unit 2	19 (19)	7 (0)	48.185	81.583	210	75°S	297.0	70.7	297.4	70.4	100	3.4
18	A6, unit 2	5 (5)	—	48.263	81.333	231	—	299.2	64.9	299.6	64.6	518	3.4
19	A6, unit 2	10 (10)	—	48.530	80.640	186	—	285.5	69.5	286.0	69.5	67	5.9
20	A6, unit 2	18 (17)	—	48.550	80.593	187	90°	316.4	72.7	317.1	72.5	146	3.0
21	A6, unit 2	4 (4) B	—	48.553	80.575	143	—	295.3	75.6	296.2	75.6	3333	1.6
		4 (4) U	—					212.9	−9.2	212.6	−8.0	150	7.5
22	A6, unit 2	5 (4)	—	48.562	80.553	161	75°S	328.8	71.6	329.3	71.3	169	7.1
23	A6, unit 2	5 (5)	—	48.888	79.505	185	—	300.2	75.3	301.5	75.5	244	4.9
24	A6, unit 2	17 (16)	—	48.918	79.418	220	75°S	292.8	68.7	293.2	69.2	39	6.0
25	A6, unit 1	4 (4)	—	49.585	77.102	148	90°	276.0	51.1	274.3	53.7	277	5.5
26	A6, unit 1	6 (6)	—	49.880	75.780	85	85°N	283.1	55.3	281.1	58.1	547	2.9
27	A7	5 (5)	—	47.075	83.745	80		146.3	−78.1	143.5	−78.0	223	5.1
28	A7	7 (7)	1 (0)	47.075	83.745	80		157.9	−83.4	151.1	−83.4	345	3.3
29	A8	5 (4)	—	48.151	79.148	~165		268.9	64.6	267.7	65.5	288	5.4
30	A8	5 (4)	—	48.214	79.977	~130		277.5	51.4	276.9	52.1	95	9.5
31	A8	5 (4)	—	48.616	78.221	~140		261.5	66.1	259.8	67.5	88	9.8
32	A8	5 (4)	—	48.676	78.126	~85		258.5	46.0	256.5	48.4	24	19.1
33	A8	5 (3)	1 (0)	~49.32	~77.18			264.5	63.5	262.8	65.6	192	8.9

NOTES: N_A, number of samples measured using AF demagnetization; n_A, number of samples utilized in mean calculation; N_T, number of samples measured using thermal demagnetization; n_T, number of samples utilized in mean calculation; B, baked host rock; U, unbaked host rock; Lat. and Long., latitude (N) and longitude (W) of the sampling site, measured to an uncertainty of less than 0.002°, D and I, mean declination and inclination of characteristic remanence; D' and I' mean declination and inclination recalculated for common reference locality of 48.0°N, 81.0°W; k, precision parameter; α_{95}, radius of circle of 95% confidence. Dyke means are given in Table 3. Sites 27 and 28 are about 200 m apart. Thickness and dip of the dyke are measured at the sampling site.

normal magnetization direction, but not both. Magnetic polarity is defined according to the convention used in paleomagnetic studies of Keweenawan rocks of the Lake Superior basin (e.g, Robertson and Fahrig 1971). These directions are considered primary on the basis of paleomagnetic field tests described below. Examples of normally magnetized specimens are illustrated in Fig. 2; examples of reversely magnetized specimens are shown in Fig. 3. At these sites, only a single component of magnetization was observed above 20 mT or ~450°C, after removal of viscous components with directions that are usually close to the present Earth's field. In both normal and reversely magnetized samples, most blocking temperatures for the characteristic component fall between about 450 and 570°C, indicative of low-Ti titanomagnetite. This result indicates that the coarse titanomagnetite grains observed in thin section cannot be the remanence carriers, since microprobe analyses from dyke A6, discussed earlier, revealed compositions of Usp_{32-74}. Therefore, the remanence must be carried by finer grained low-Ti titanomagnetite whose distribution is presently unresolved.

Dyke A6 has been interpreted to consist of two pulses on the basis of petrology, geochemistry, and paleomagnetism. Petrology and geochemistry indicate that an earlier pulse, labelled unit 1, was emplaced along the entire length of the dyke, and a later pulse, unit 2, was emplaced in the interior over at least a third of the dyke length between sites 16 and 24, inclusive. Paleomagnetic study revealed two characteristic directions of magnetization which distinctly differ by about 20° (Tables 2 and 3). Samples from sites 13−15, 25, and 26 (Fig. 1) of unit 1 exhibit only one of the two observed directions, which is called the unit 1 magnetization direction. The second direction is seen in samples of sites 16−20 and 22−24 from both units 1 and 2. This second remanence direction, called the unit 2 magnetization direction, was also observed in host-rock samples at site 21 where the dyke itself is not exposed. It is proposed that both unit 1 samples and adjacent host rock at sites 16−24 were overprinted during the subsequent intrusion and cooling of unit 2. Since the geographic distribution of sites exhibiting the two magnetization directions is

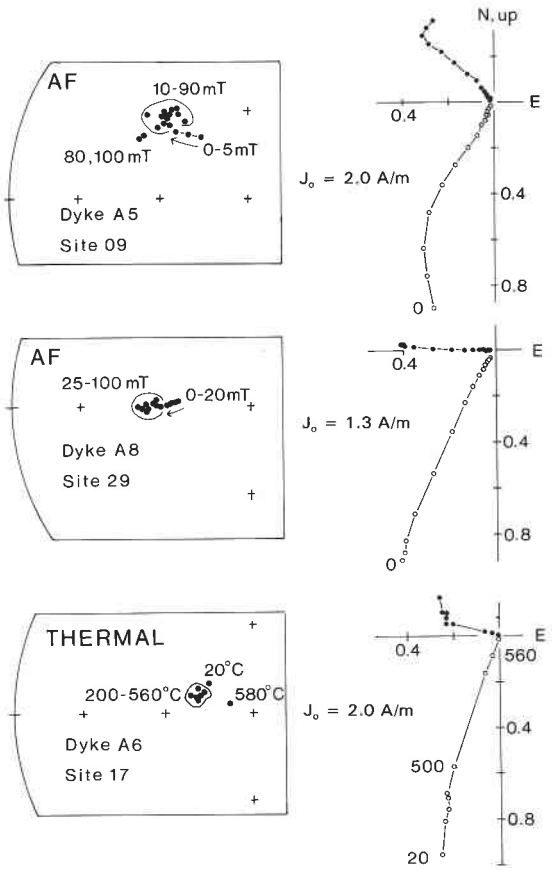


FIG. 2. Typical paleomagnetic results for specimens from normal-polarity Abitibi dykes during alternating-field (AF) or thermal demagnetization. Directional data are plotted with respect to present horizontal on an equal-area stereonet with closed (open) symbols indicating down (up) directions. On orthogonal component plots closed symbols indicate projection onto the horizontal plane and open symbols indicate projection onto the up and east vertical plane. J_o, NRM intensity. Dyke and site locations are given in Fig. 1.

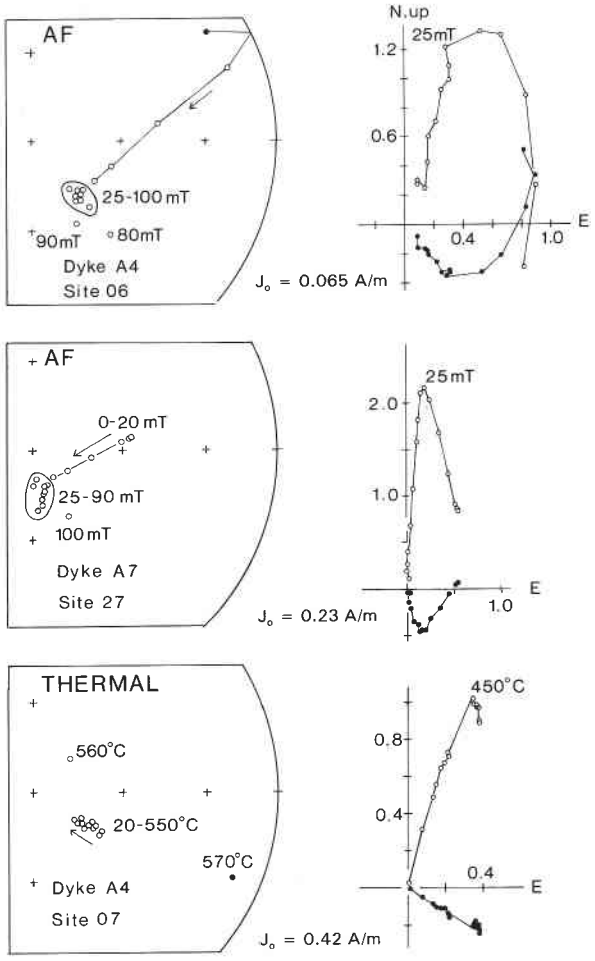


FIG. 3. Typical paleomagnetic results for specimens from reversed polarity Abitibi dykes. Symbols and conventions as in Fig. 2.

consistent with the geochemically determined distribution of the two units, we consider both remanences to be primary thermoremanent magnetizations acquired during the cooling of units 1 and 2 at different times. The units 1 and 2 magnetizations are considered as distinct "dykes" for the purpose of the statistical calculation of a mean direction for the swarm.

The five sites from dykes A1, A2, and A3 yielded more complicated results. The single site (site 02) collected in dyke A2 carries very scattered and unstable directions, and no characteristic direction of magnetization could be determined for this site.

Alternating-field demagnetization of samples from sites 03 and 04 from dyke A3 (Fig. 1) revealed normal polarity directions with substantially shallower inclinations than observed in other normally magnetized sites in the swarm. A few specimens at these sites showed anomalous swings, hinting at the presence of a reversely magnetized component. Therefore, we conducted detailed thermal demagnetization experiments on 24 samples which successfully resolved three stable components of magnetization and a viscous component, each occupying a discrete blocking temperature spectrum (Figs. 4 and 5). Every sample from these two sites showed the same pattern. A component of viscous remanent magnetization (VRM) (Fig. 4) was eliminated between room temperature and 250°C. Next, a normal component with northwest declination and intermediate inclination, N_K (Figs. 4 and 5), was removed between 250 and 475°C. Then, a reversed component of southeast declination and steep inclination, R_K, was removed between 475 and 510°C. Finally, a distinct normal component of northwest declination and steep inclination, N, was removed between 510 and about 580°C, the Curie temperature of magnetite. The removal of the reversed R_K component resulted in a dramatic increase in the remanent intensity upon thermal demagnetization (Fig. 4), with the peak intensity located at the R_K to N transition temperature. The high blocking temperature component, N, is indistinguishable from the characteristic primary magnetization of other stable Abitibi dyke sites and, hence, is interpreted to be the primary component at these sites as well. Since the dyke will have cooled following intrusion in a short time relative to that necessary for a field reversal, the lower temperature components, R_K and N_K, are thought to be secondary overprints. The fact that N, R_K, and N_K occupy discrete but adjacent blocking temperature spectra suggests that N_K and R_K were acquired as thermal overprints during the cooling that followed a mild reheating event or events. The poles for overprints R_K and N_K are 49.9°N, 158.2°W (dm = 8.3°, dp = 6.5°, 5 samples) and 35.7°N, 178.8°W (dm = 5.2, dp = 3.2°, 12 samples), respectively. The pole for the characteristic Abitibi dyke remanence, N, of this dyke is given in Table 2.

In addition to the clear overprints recorded at sites 03 and

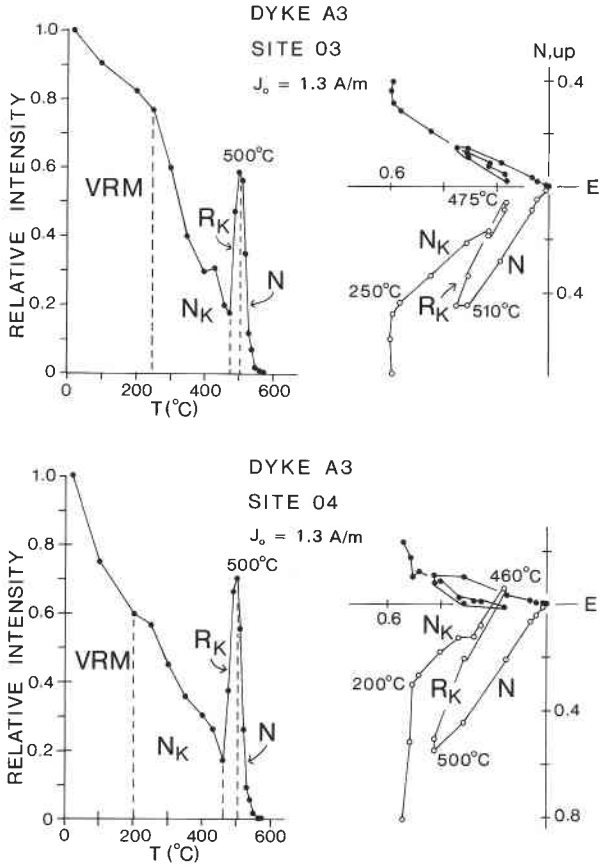


FIG. 4. Thermal demagnetization results for typical specimens from sites 03 and 04 of dyke A3. Blocking temperature ranges on the intensity decay plots are labelled VRM, N_K, R_K, and N and correspond to elimination of magnetization components with distinct directions as isolated on the orthogonal component plots. Symbols as in Fig. 2.

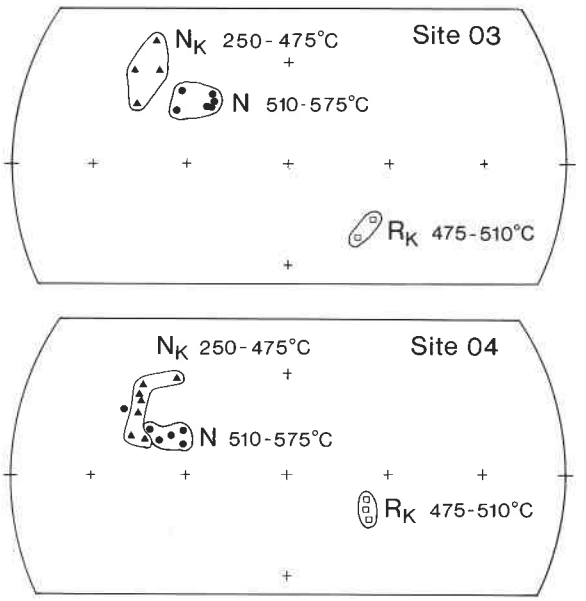


FIG. 5. Summary of magnetization components N_K, R_K, and N upon thermal demagnetization for individual samples at sites 03 and 04 of dyke A3. Data are plotted on equal-area stereonets. Symbols as in Fig. 2.

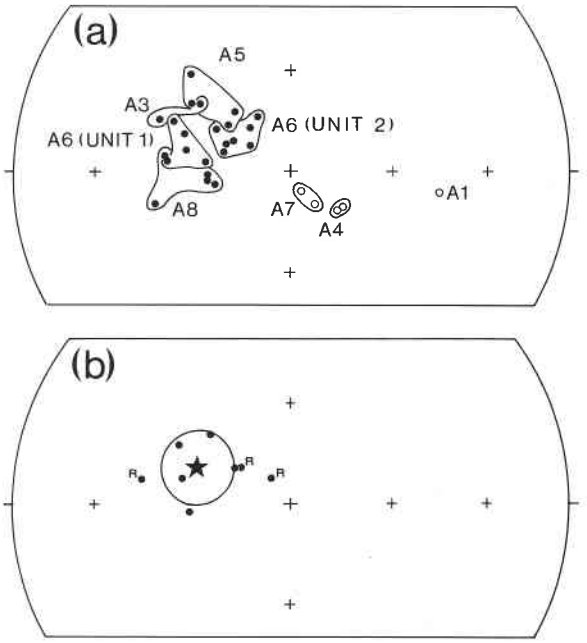


FIG. 6. Characteristic paleomagnetic directions for the Abitibi dyke swarm. To facilitate comparison of directions from widely separated geographic localities, directions are converted to a common reference site at 48°N, 81°W. (*a*) Mean site directions grouped and labelled by dyke. Closed (open) symbols indicate normal (reversed) remanence directions. (*b*) Mean dyke directions. R, reversed-polarity dykes plotted at their normal-polarity antipodes; ★, mean direction for the normally magnetized dykes, with its 95% circle of confidence.

04, three additional sites (sites 05, 09, and 17) show evidence of similar secondary components. In most of the samples from these sites, the intensity of remanence exhibits a distinctive peak upon thermal demagnetization in the range of 450–530°C, suggesting elimination of the reversed overprint component, R_K. Despite extensive thermal demagnetization experiments at site 05 (along the probable extension of dyke A3), clear separation of the overprint and the characteristic components proved impossible. Therefore, the data from site 05 could not be used in the calculation of the mean Abitibi direction. Finally, at sites 09 and 17, too few thermal demagnetization steps were done in the narrow transition interval to attempt resolution of the reversed overprint component.

The data from the single site (site 01) of dyke A1 are difficult to interpret. Ten samples were collected. Of these, four samples carry a reversed-polarity direction that is somewhat shallower than the direction observed in reversely magnetized dykes A4 and A7. Two additional samples carry directions that may represent normal-polarity Abitibi components, and the four other samples have stable but scattered remanence directions. In light of the complexity of the results at this site, the calculated remanence direction in Table 2, which is based only on the four reversely magnetized samples, should be used with caution.

Mean characteristic remanence directions for all usable sites are plotted in Fig. 6*a* and listed in Table 2. Clusters of site means for a single dyke are identified in Fig. 6*a*, and dyke mean directions are shown in Fig. 6*b* and listed in Table 3. The mean magnetization direction for normal-polarity Abitibi dykes is as follows: declination $D = 291.5°$, inclination $I = 60.9°$, precision parameter $k = 53$, radius of cone of 95% confidence $\alpha_{95} = 10.6°$, number of sites $N = 23$ from five

TABLE 3. Summary of mean directions for Abitibi dykes

Dyke	N (n)	D' (°)	I' (°)	k	α_{95} (°)	Lat. (°)	Long. (°)	dm (°)	dp (°)
A1	1 (1)	99.3	−45.2	—	—	25.5	158.6	—	—
A2	1 (0)	—	—	—	—	—	—	—	—
A3	3 (2)	298.3	52.4	—	—	42.1	165.6	—	—
A4	2 (2)	126.4	−71.5	—	—	57.0	136.2	—	—
A5	5 (4)	310.5	59.1	72	10.9	54.1	165.3	16.3	12.2
A6, unit 1	5 (5)	283.1	57.2	153	6.2	35.2	151.2	9.1	6.6
A6, unit 2	9 (7)	302.7	70.9	189	4.4	54.8	137.1	7.7	6.7
A7	2 (2)	146.2	−80.7	—	—	61.5	102.3	—	—
A8	5 (5)	264.9	60.1	68	9.4	26.3	138.0	14.2	10.7
Mean (normal dykes)	5 (5)	291.5	60.9	53	10.6	42.8	151.5	16.3	12.5
Mean (all dykes)	8 (8)	291.9	63.1	33	9.8	44.4	148.6	15.4	12.1

NOTES: N, number of sites measured (Table 2); n, number of sites utilized in mean calculation; D' and I', mean declination and inclination calculated for common reference locality of 48.0°N, 81.0°W; k, precision parameter; α_{95}, radius of circle of 95% confidence; Lat. and Long., latitude (N) and longitude (W) of the mean paleopole; dm and dp, semi-axes of the oval of 95% confidence. Sites 2, 5, and 8 (Table 2) are not stably magnetized and were not included in dyke statistics. Baked host rock samples were not used in the mean calculations, and therefore site 21, with no dykes samples, was not used in mean calculations. Site 16, which contains 4 baked host and only 1 dyke sample, was also not used.

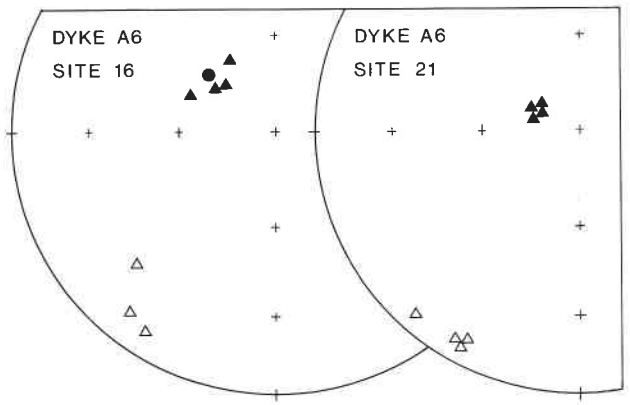


FIG. 7. Paleomagnetic data for sites 16 and 21 of dyke A6. ●, steep-down magnetization direction obtained from an Abitibi dyke sample; ▲, steep-down magnetization direction of baked Matachewan dyke host rock samples; △, shallow-up direction of unbaked Matachewan dyke host rock samples.

dykes. The corresponding pole is located at 42.8°N, 151.5°W, dm = 16.3°, dp = 12.5°. The reversely magnetized data are not sufficiently well constrained in the present study to allow the calculation of an independent reversed pole, although it might be noted that a mean pole based on both normal and reversed data is not distinguishable from that based on normal data alone (Table 3).

Paleomagnetic field tests for remanence age

Paleomagnetic field tests were conducted to test whether the normal and reversed magnetizations that characterize the Abitibi swarm are primary or secondary. Baked contact, secular variation, and reversal-age correlation tests are described below.

Baked contact tests were attempted at sites 16 and 21 (Fig. 1). At each locality the host rock, consisting of north-trending diabase dykes of the 2452 Ma (Heaman and Tarney 1989) Matachewan swarm (Halls 1991), was sampled both close to and far from Abitibi dyke A6 (Fig. 7). At site 16 the host-rock samples adjacent to the Abitibi dyke contact carry a unit 2 Abitibi direction, indistinguishable from that of the single sample collected in the chilled margin of the Abitibi dyke itself, whereas the host-rock samples at a distance of 240 m from the contact carry the typical Matachewan dyke direction reported by Halls (1991). This constitutes a positive baked contact test (Everitt and Clegg 1962) and demonstrates that the unit 2 remanence of this Abitibi dyke was acquired at the time of unit 2 intrusion and cooling. At site 21 the dyke itself does not outcrop, but its position and approximate 150 m width are evident from its prominent aeromagnetic signature. The dyke does outcrop 1 km to the east (site 20) and 6 km to the west (site 22) of site 21 where it carries the unit 2 Abitibi magnetization. At site 21, samples of the host rock 50 m away from the Abitibi dyke contact carry the unit 2 Abitibi remanence, but those from a distance of 370 m from the contact carry a primary Matachewan dyke direction. Although not a baked contact test according to the criteria proposed by Buchan and Halls (1990) because of the lack of dyke material at the sampling locality itself, this data does support the interpretation that the unit 2 remanence is primary.

The primary nature of the unit 1 direction of dyke A6 was not tested directly by a baked contact test, but the absence of a chilled contact between units 1 and 2 indicates that a short interval, perhaps only a few hundred years, separates their emplacement based on calculated cooling rates (e.g., Delaney 1987). Given this and the observation that the distribution of units 1 and 2 remanences correlates with the geochemically defined distribution of the two units, we conclude that the unit 1 remanence direction must also be primary.

The primary nature of the magnetic remanence of the other Abitibi dykes can be assessed using secular variation and reversal-age correlation tests. These paleomagnetic field tests were described but not named in Halls (1986) and Buchan and Halls (1990). As noted above, each stably magnetized dyke of the Abitibi swarm exhibits a consistent characteristic remanence direction along its length (Fig. 6), even though the directions differ slightly between dykes. Because the blocking temperature spectra are consistent from one dyke to the next, this constitutes a positive secular variation test of the primary nature of the characteristic remanence. The consistency of directions

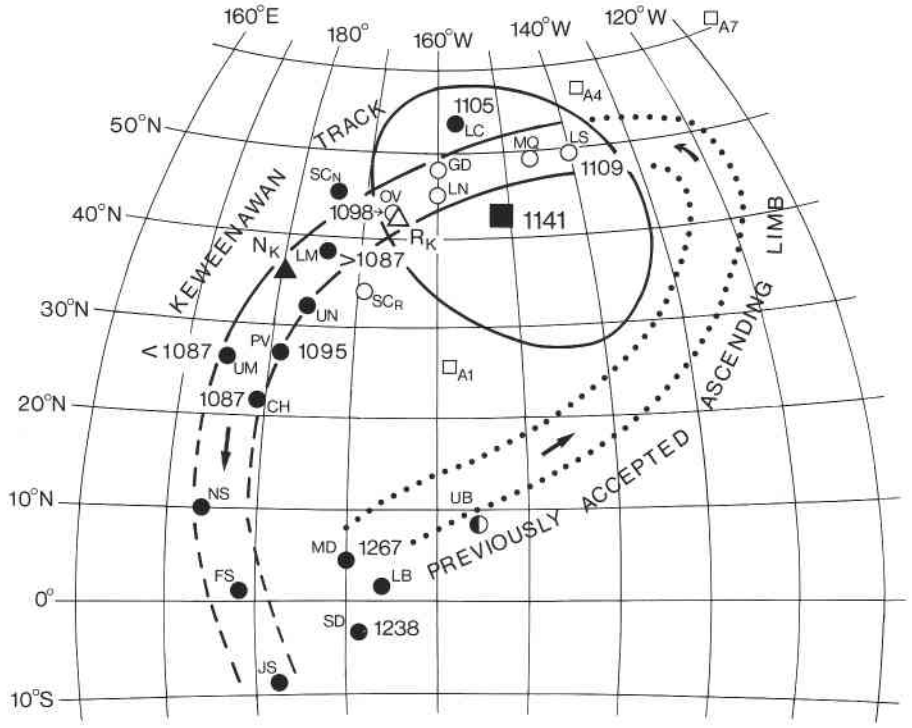


FIG. 8. Revised Abitibi dyke paleomagnetic poles compared with selected poles along the Logan Loop. The location of the ascending limb of the Logan Loop (dotted) follows Green et al. (1987). ● and ○, normal- and reversed-polarity poles; ◐, pole for a unit that exhibits a nearly symmetric field reversal; ■, mean pole, with its oval of 95% confidence, for normally magnetized Abitibi dykes of this study; □, mean directions for reversely magnetized Abitibi dykes (A1, A4, and A7); △ and ▲, means of reversed (R_K) and normal (N_K) Keweenawan overprint poles from sites 03 and 04 of Abitibi dyke A3. Other poles are labelled as follows: Copper Harbour conglomerate (CH), Freda sandstone (FS), Grand Portage dykes (GD), Jacobsville sandstone (JS), Lackner Lake complex (LC), Logan sills (LS), Lower Borden sequence (LB), Lower Michipicoten volcanics (LM), Lower North Shore volcanics (LN), Mackenzie dykes (MD), Marquette dykes (MQ), Nonesuch shale (NS), Osler volcanics (OV), Portage Lake volcanics (PV), Shenango complex normal (SC_N), Shenango complex reversed (SC_R), Sudbury dykes (SD), Upper Borden sequence (UB), Upper Michipicoten volcanics (UM), and Upper North Shore volcanics (UN). The sources for these poles are Fahrig et al. (1981), Halls and Pesonen (1982), Green et al. (1987), Symons (1989), Buchan and Halls (1990), and Constanzo-Alvarez et al. (1993). Complete summaries of Keweenawan poles can be found in Symons et al. (1994) and Halls and Pesonen (1982). High-precision U−Pb ages for key poles are given in millions of years (Ma) and are from Davis and Sutcliffe (1985), Davis and Paces (1990), and Heaman and Machado (1992). The age for the PV is the average of 1096.2 ± 1.8 and 1094.0 ± 1.5 Ma for the bottom and top of the sequence, respectively (Davis and Paces 1990).

along individual dykes suggests that each dyke was emplaced and cooled in a short period of time relative to secular variation of the Earth's magnetic field. Similarly, the presence of adjacent Abitibi dykes that are consistently of either reversed or normal polarity, particularly dykes A4 and A5, constitutes a positive reversal-age correlation test and therefore suggests that the remanence is primary.

Discussion

The normal-polarity Abitibi pole (42.8°N, 151.5°W, dm = 16.3°, dp = 12.5°) determined in this study plots near the northern end of the Logan Loop (Fig. 8) of the North American polar wander path (e.g., Robertson and Fahrig 1971). The nature of the loop, based largely on data from Keweenawan rocks of the Lake Superior region, has long been controversial because of a paucity of data on the older ascending limb and evidence of asymmetry in poles defining the younger descending limb (e.g., Halls and Pesonen 1982; Pesonen and Halls 1983; Symons et al. 1994). Paleomagnetic data of somewhat similar age have also been described from the Grand Canyon Supergroup of Arizona (e.g., Elston and Gromme 1974; summarized in Link et al. 1993), but will not be discussed here because they have yet to be published in detail and age constraints are poor.

The Logan Loop

The older ascending limb of the Logan Loop (Fig. 8) is only constrained at either end. At the older end are the primary poles for the 1267 Ma Mackenzie dyke swarm (Fahrig and Jones 1969; LeCheminant and Heaman 1989), the 1238 Ma Sudbury dyke swarm (Palmer et al. 1977; Krogh et al. 1987), and sediments and volcanic rocks of the Borden Basin (Fahrig et al. 1981). The lower units of the Borden Basin (pole LB in Fig. 8), the Nauyat volcanics and Adams Sound Formation, are thought to be coeval with Mackenzie and Sudbury dykes, since all three poles LB, MD, and SD have similar directions and exclusive normal polarity (Fahrig et al. 1981). In contrast, the stratigraphically younger Strathcona Sound Formation at the top of the Borden sequence exhibits six symmetric magnetic reversals and has a pole position (UB) that is close to but is slightly northeast of the MD, SD, and LB grouping (Fig. 8).

At the younger end of the ascending limb of the Logan Loop (Fig. 8) is the 1109 Ma (U−Pb; Davis and Sutcliffe 1985) Logan sill pole (Halls and Pesonen 1982) from the Keweenawan basin. Other poles that were previously associated with the

ascending limb of the loop have been reinterpreted. Earlier published poles for the "Abitibi" dykes (e.g., Fahrig et al. 1965; Larochelle 1966; Irving and Naldrett 1977) fall along the ascending arm of the Loop, but are now known to have been calculated from data from both Abitibi and 2167 Ma Biscotasing dykes (Buchan et al. 1993*a*) and should not be used. Although the poles for the Powder Mill group of the Keweenawan basin were originally plotted on the ascending limb, reevaluation of the data suggests both a different age and tectonic correction which move the Powder Mill poles onto the descending limb (Palmer and Halls 1986). The pole for the Aillik dykes of the Nain Province (Fahrig and Larochelle 1972) falls between that of the Mackenzie dykes and Logan sills. Unfortunately, however, the age of the Aillik swarm is very poorly constrained (Fahrig and Larochelle 1972), and it is not certain that the pole is primary.

The younger descending limb of the Logan Loop (Fig. 8) is well defined by poles derived from rocks associated with the Keweenawan rift system. Over the past few years most authors (e.g., Pesonen and Halls 1983; Davis and Paces 1990) have attributed a systematic separation of normal- and reversed-polarity poles along this Keweenawan Track to asymmetry in the Earth's magnetic field during the Keweenawan period. If this were so, the apparent depth of the Logan Loop would be largely an artifact of the asymmetry (Buchan and Halls 1990). However, more recently, others (e.g., Symons et al. 1994; Costanzo-Alvarez et al. 1993) have proposed that the pole separation is the result of rapid polar wander, because roughly symmetric reversals have been observed in a number of Keweenawan carbonatite complexes (e.g. Nemogescenda and Shenago Lake complexes; Symons and Garber 1974; Lewchuk and Symons 1990; Symons et al. 1994; Costanzo-Alvarez et al. 1993). In addition, the normal pole of the 1105 Ma (U–Pb; Heaman and Machado 1992) Lackner Lake complex falls among reversely magnetized Keweenawan poles (Fig. 8) (Symons 1989).

At the early end of the Logan Loop polarity asymmetry is clearly untenable. In the Upper Borden Basin sequence, the Strathcona Sound Formation exhibits six reversals, all of which are nearly symmetric (Fahrig et al. 1981). Thus any asymmetry recorded by Keweenawan units can have begun no earlier than deposition of the Strathcona Sound Formation, which is probably younger than the 1238 Ma age of the Sudbury dykes.

A key piece of evidence that has been used to argue for asymmetry during the Keweenawan period is a sequence of reversed–normal–reversed–normal paleomagnetic directions in volcanic flows at Mamainse Point on the eastern shore of Lake Superior, which has been interpreted to show a distinct separation based on polarity (Palmer 1970; Robertson 1973). Geochemical studies of the Mamainse Point volcanics (Klewin and Berg 1990) demonstrate that asymmetry cannot be due simply to fault repetition as originally proposed by Palmer (1970) and Robertson (1973). Our reexamination of the paleomagnetic data in Palmer (1970) and Robertson (1973), however, indicates that the magnetic direction from the upper reversed flows is statistically distinct from the direction of the lower reversed flows. However, at least in the case of Robertson (1973), the magnetic direction from the upper reversed flows is indistinguishable from the directions of both upper and lower normal flows. Furthermore, the mean direction for the upper reversed flows at Mamainse Point is based on only five sites of Palmer (1970) and two of Robertson (1973) and may not average secular variation.

Thus, the paleomagnetic data at Mamainse Point do not necessarily require reversal asymmetry. They are consistent with rapid apparent polar wander along the Keweenawan Track (Fig. 8) following extrusion of the lower reversed flows, with little polar wander during the rest of the normal–reversed–normal sequence.

Although other sequences of Keweenawan volcanic flows also show a similar separation of reversed and normal poles, in each case only a single reversal is observed (e.g., Pesonen and Halls 1983). In most cases a conglomerate–sandstone unit is observed between the reversed- and normal-polarity sequences (e.g., Halls 1974), indicating a break in volcanism which may mark a hiatus. Recent precise U–Pb dating (Davis and Paces 1990) has shown that the transition from dominantly reversed to dominantly normal Keweenawan poles occurred between 1097.6 ± 3.7 Ma (Davis and Sutcliffe 1985) and 1096.2 ± 1.8 Ma (Davis and Paces 1990), the ages of the upper portion of the reversely magnetized Osler volcanic sequence and the normally magnetized Portage Lake volcanic sequence, respectively. The tight age constraints could be interpreted to indicate that up to 15–20° of apparent movement along the Keweenawan Track can be ascribed to asymmetry (Buchan and Halls 1990). However, if the uncertainties in the radiometric ages (Davis and Paces 1990) and the uncertainties in the poles (Halls and Pesonen 1982; Halls 1974) of the Portage Lake and reversely magnetized Osler volcanics are taken into account, this data could also represent rapid plate movement at a minimum rate of about 20 cm/year, corresponding to 1500 km of apparent pole movement in 7 Ma.

If the above interpretation is correct, then the Keweenawan Track includes an older reversed-polarity interval with one or more minor normal periods to account for the Lackner Lake normal pole as proposed by Symons et al. (1994). This is followed by rapid apparent polar wander to a dominantly normal portion of the track with one or more reversed intervals to account for the upper Mamainse Point reversed flows. The apparent polar wander from dominantly reversed to dominantly normal polarities would correspond to plate movement from high to low paleolatitudes. Paleointensity data are consistent with such a paleolatitudinal shift (Pesonen and Halls 1983).

Implications of the revised Abitibi pole

The mean pole for the normally and reversely magnetized Abitibi dykes does not differ significantly from the mean pole based on normal dykes alone (Table 3). With only three reversed dykes, two with directions steeper than the normal polarity mean and one that is shallower (Fig. 6), the present data set is insufficient to test the asymmetry model at 1141 Ma, although such a test would be possible if more reversely magnetized dykes could be located.

Nevertheless, the normal-polarity pole for the Abitibi swarm is reasonably well determined and is located at the northern end of the Logan Loop (Fig. 8) where it establishes a minimum depth of about 40° for the loop when measured from the Sudbury dyke pole (SD) regardless of whether the Keweenawan data reflect an asymmetric reversal or not. No data are available to constrain the polar wander path between 1238 and 1141 Ma, the ages of Sudbury and Abitibi dyke poles, respectively, nor between 1141 and 1109 Ma, the ages of the Abitibi dyke and Logan sill poles, respectively. Indeed, it is not inconceivable that the Mackenzie and Sudbury dyke poles should be plotted in the opposite hemisphere relative to the Keweenawan Track, such that the apparent polar wander path tracks over the

geographic pole between Mackenzie dyke and Keweenawan times rather than forming a closed loop (e.g., Morris et al. 1979; Baer 1979).

Keweenawan overprint on some Abitibi dykes

Reversed and normal pole positions that correspond to the secondary magnetization components, R_K and N_K (Fig. 4 and 5), at sites 03 and 04 of dyke A3 near Wawa (Fig. 1) plot along the Logan Loop in the reversed and normal Keweenawan groupings, respectively (Fig. 8). They also occupy discrete blocking temperature ranges of 475–510 and 250–475°C for the reversed and normal overprints, respectively, immediately below the blocking temperatures of the characteristic Abitibi dyke magnetization at these sites (Fig. 4). Sites 03 and 04 are located adjacent to the Keweenawan rift in the eastern Lake Superior region. Therefore, we interpret these components to have been acquired as thermal overprints during the reversed and normal Keweenawan periods between about 1100 and 1090 Ma about 40–50 million years after Abitibi dyke emplacement. In support of this explanation, we note that small Keweenawan volcanic outliers have been mapped about 40 km southwest of site 03 (Reid et al. 1991, p. 62; Reid et al. 1992*a*, 1992*b*).

The extent of this Keweenawan overprinting is not well defined, perhaps because most Abitibi dyke samples from sites other than 03 and 04 do not have blocking temperatures in the appropriate range. However, evidence for a weak secondary magnetization was noted at sites 05, 09, and 17, up to about 300 km from the Keweenawan rift. On the other hand, Keweenawan overprints have not been reported in other studies of pre-Keweenawan units near Wawa (Vandall and Symons 1990). Once again, this may be due to the absence of appropriate blocking temperatures. Another study that examined diabase dykes about 100 km farther south (Halls and Shaw 1988) is inconclusive regarding the presence of an overprint because no thermal demagnetization results were reported.

Conclusions

Dykes of the 1141 Ma northeast-trending Abitibi swarm, composed of olivine gabbro to olivine-bearing monzogabbro, have a primary paleomagnetic pole located at 42.8°N, 151.5°W, dm = 16.3°, dp = 12.5° based on normal polarity data from 23 sites in five dykes (four dykes and a second pulse in one of the dykes). Previously published poles that purport to represent the Abitibi swarm should no longer be used because they are now known to be derived in part from the 2167 Ma quartz tholeiite Biscotasing (formerly Preissac) swarm.

The revised normal-polarity pole for the Abitibi dyke swarm implies a minimum depth for the Logan Loop of 40°. Reversed polarity Abitibi dykes have been identified in this study for the first time, but are too few in number to average out secular variation and hence to test for reversal asymmetry. However, the available data, particularly recent data from carbonatite complexes (e.g., Symons et al. 1994), suggests that rapid polar wander rather than asymmetry can account for the general separation of reversed- and normal-polarity Keweenawan poles.

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Walter Fahrig, Ken Card, and Henry Halls provided much useful discussion on distinguishing the various northeast-trending dyke swarms of the southern Superior Province. Ike Osmani, John Percival, and Ron Sage were helpful in suggesting Abitibi dyke sampling localities. R.E.E. acknowledges the field assistance of Dave Sloski and the late Jacques Bisson. Matt Pyne conducted the thermal demagnetization experiments on samples from site 17. Modifications of the thermal demagnetization unit (discussed in the text) were done by Garry Freda in the Geological Survey of Canada paleomagnetic laboratory (Ottawa). Reviews by Dave Dunlop and Dave Symons are gratefully acknowledged.

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